



SE507 - SOFTWARE TESTING AND EVALUATION

A Comprehensive Testing Strategy for a Library Management System

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Functional Testing

Functional testing is a type of black-box testing that verifies the system's compliance with its specified functional requirements, as outlined in the Software Requirements Specification (SRS) document. This chapter details the functional test cases executed to ensure that each feature of the Library Management System behaves as expected from an end-user perspective, encompassing various aspects such as user interactions, data manipulations, and system responses.

The functional test cases are categorized into the following sub-sections:

- Authentication and Authorization Testing
- Entity Management (CRUD Operations) Testing
- Specific Feature Testing
- Use Case Testing
- State Transition Testing

Each sub-section includes a brief overview and references the detailed test case documents located in the `test_artifacts/test_cases/I_Functional_Testing/` directory.

1.1 Authentication and Authorization Testing

This section focuses on verifying that user authentication (login, logout, registration) and role-based access control (RBAC) mechanisms function correctly. It ensures that only authorized users can access specific features and that security measures related to user sessions are properly enforced. The test cases cover successful logins for both Librarians and Members, various failed login scenarios, successful logouts, and verification of access restrictions for different user roles.

The detailed test cases for Authentication and Authorization are provided in:

Table 1.1: Authentication and Authorization Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Sub-Section: User Registration (by Librarian)			
			Continued on next page

Table 1.1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_AUTH_REG_001	Successful new user (Member) registration by Librarian	• Login as Librarian. • Navigate to 'User Management' or 'Add User' section. • Fill in all required fields with valid and unique data for a new 'Member' (e.g., email, password, full name). • Submit the registration form.	• System accepts the data. • A success message "User registered successfully" (or similar) is displayed. • The new user appears in the list of users. • The new user can subsequently log in (to be tested in Login cases).
TC_AUTH_REG_002	Attempt to register a new user with an existing email	• Login as Librarian. • Navigate to 'Add User' section. • Fill in user details, using an email address that already exists in the system. • Submit the registration form.	• System rejects the registration. • An appropriate error message is displayed (e.g., "Email already exists"). • The user is not created.
TC_AUTH_REG_003	Attempt to register a new user with missing required fields	• Login as Librarian. • Navigate to 'Add User' section. • Fill in user details but leave one or more mandatory fields blank (e.g., password or email). • Submit the registration form.	• System rejects the registration. • An appropriate error message is displayed, indicating which fields are required. • The user is not created.
TC_AUTH_REG_004	Attempt to register a new user with invalid data format (e.g., email)	• Login as Librarian. • Navigate to 'Add User' section. • Fill in user details, providing an incorrectly formatted email address (e.g., "usertest.com"). • Submit the registration form.	• System rejects the registration. • An appropriate error message is displayed (e.g., "Invalid email format"). • The user is not created.
Sub-Section: User Login			
Continued on next page			

Table 1.1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_AUTH_LOGIN_001	Successful login with valid Librarian credentials	• Navigate to the login page. • Enter valid Librarian email. • Enter corresponding valid password. • Click the 'Login' button.	• User is authenticated successfully. • User is redirected to the Librarian dashboard/homepage. • Librarian-specific functionalities are visible/accessible.
TC_AUTH_LOGIN_002	Successful login with valid Member credentials	• Navigate to the login page. • Enter valid Member email. • Enter corresponding valid password. • Click the 'Login' button.	• User is authenticated successfully. • User is redirected to the Member dashboard/homepage. • Member-specific functionalities are visible/accessible.
TC_AUTH_LOGIN_003	Attempt login with invalid email	• Navigate to the login page. • Enter an email that does not exist. • Enter any password. • Click the 'Login' button.	• Login fails. • An appropriate error message is displayed (e.g., "Invalid email or password"). • User remains on the login page.
TC_AUTH_LOGIN_004	Attempt login with valid email but invalid password	• Navigate to the login page. • Enter a valid existing email. • Enter an incorrect password for that email. • Click the 'Login' button.	• Login fails. • An appropriate error message is displayed (e.g., "Invalid email or password"). • User remains on the login page.
TC_AUTH_LOGIN_005	Attempt login with empty email field	• Navigate to the login page. • Leave the email field empty. • Enter any password. • Click the 'Login' button.	• Login fails. • An appropriate error message is displayed (e.g., "Email is required"). • User remains on the login page.
Continued on next page			

Table 1.1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_AUTH_LOGIN_006	Attempt login with empty password field	• Navigate to the login page. • Enter a valid email. • Leave the password field empty. • Click the 'Login' button.	• Login fails. • An appropriate error message is displayed (e.g., "Password is required"). • User remains on the login page.
Sub-Section: User Logout			
TC_AUTH_LOGOUT_005	Successful logout for a logged-in Librarian	• Login as Librarian. • Click the 'Logout' button/link.	• User is logged out successfully. • User is redirected to the login page or public homepage. • Session is terminated (user cannot access protected pages without logging in again).
TC_AUTH_LOGOUT_006	Successful logout for a logged-in Member	• Login as Member. • Click the 'Logout' button/link.	• User is logged out successfully. • User is redirected to the login page or public homepage. • Session is terminated.
Sub-Section: Role-Based Access Control (RBAC)			
TC_AUTH_RBAC_001	Verify Librarian can access Librarian-specific features (e.g., User Management)	• Login as Librarian. • Attempt to navigate to 'User Management' section. • Attempt to navigate to 'Book Management' (add/edit/delete books).	• Librarian successfully accesses 'User Management' page. • Librarian can view and interact with user administration tools. • Librarian successfully accesses 'Book Management' page and can perform CRUD operations on books.
Continued on next page			

Table 1.1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_AUTH_RBAC_002	Verify Member cannot access Librarian-specific features (e.g., User Management)	• Login as Member. • Check if 'User Management' link/button is visible. • Attempt to navigate directly to the User Management URL (if known). • Check if 'Add Book' or 'Edit Book' options are available.	• 'User Management' link/button is not visible or is disabled. • Direct navigation to User Management URL redirects to an error page or member dashboard. • 'Add Book'/'Edit Book' options are not available. Access is denied.
TC_AUTH_RBAC_003	Verify Member can access Member-specific features (e.g., Search Books, View Borrowal History)	• Login as Member. • Attempt to navigate to 'Search Books' page. • Attempt to view 'My Borrowal History'.	• Member successfully accesses 'Search Books' page and can search. • Member successfully accesses 'My Borrowal History' and can view their own history.
TC_AUTH_RBAC_004	Verify guest (not logged in) user access to public pages and restriction from protected pages	• Ensure no user is logged in (clear session/cookies or use incognito window). • Attempt to navigate to the login page. • Attempt to navigate to a public book search page (if applicable). • Attempt to navigate to 'User Management' URL. • Attempt to navigate to 'My Borrowal History' URL.	• Login page is accessible. • Public book search page is accessible (if such a feature exists for guests). • Access to 'User Management' URL is denied, likely redirecting to login page. • Access to 'My Borrowal History' URL is denied, likely redirecting to login page.

1.2 Entity Management (CRUD Operations) Testing

This section covers the Create, Read, Update, and Delete (CRUD) operations for the core entities of the Library Management System, including Books, Authors, Genres, Users, Borrowals, and Reviews. These tests ensure that data can be correctly managed by librarians and that members have appropriate viewing and interaction capabilities, while also verifying referential integrity constraints.

1.2.1 Book Management

Test cases for Book Management verify the addition, viewing, updating, and deletion of book records, including scenarios where deletion is prevented due to active borrowals.

1.2.2 Author Management

This covers the creation and deletion of author records, including checks for referential integrity when an author is linked to existing books.

1.2.3 Genre Management

Similar to Author Management, these test cases validate the creation and deletion of genre records, ensuring that genres linked to books cannot be deleted.

1.2.4 User Management

This sub-section tests the Librarian's ability to create, view, update, and delete user accounts (both Librarians and Members), and ensures that members cannot access user management functionalities or delete users with active borrowals.

1.2.5 Borrowal Management

Test cases for Borrowal Management verify the process of borrowing and returning books, including scenarios for borrowing unavailable books and ensuring members can only view their own borrowal history, while librarians can view all records.

1.2.6 Review Management

As a new feature, this section validates the ability of members to add reviews for books and librarians to manage (view and delete) all reviews.

The detailed test cases for Entity Management are provided in:

Table 1.2: Entity Management Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Sub-Section: Book Management (SRS 3.3)			
TC_BOOK_CREATE_001	Successful new book creation by Librarian	• Login as Librarian. • Navigate to 'Book Management'. • Click 'New Book'. • Fill all required fields (Name, ISBN) with valid, unique data. Select existing Author and Genre. • Submit the form.	• System accepts the data. • A success message is displayed. • The new book appears in the book list.
Continued on next page			

Table 1.2 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_BOOK_CREATE_001	Attempt to create a book with missing required fields	• Login as Librarian. • Navigate to 'Book Management'. • Click 'New Book'. • Leave the 'Name' or 'ISBN' field blank. • Submit the form.	• System rejects the creation. • An error message indicating required fields is displayed. • The book is not created.
TC_BOOK_CREATE_003	(RBAC) Verify Member cannot create a book	• Login as a Member. • Navigate to 'Book Management'.	• The 'New Book' button/option is not visible or is disabled. • Direct access to the create book API endpoint (e.g., POST /api/book/add) should be denied.
TC_BOOK_READ_001	Verify all users can view the list of books	• Login as Librarian. Navigate to 'Book Management'. • Logout. Login as Member. Navigate to 'Book Management'.	• Both Librarian and Member can successfully view the list of all books.
TC_BOOK_UPDATE_001	Successful book update by Librarian	• Login as Librarian. • Navigate to 'Book Management'. • Select a book and choose to edit it. • Modify a field (e.g., the summary). • Save the changes.	• System accepts the changes. • A success message is displayed. • The book list/details reflect the updated information.
TC_BOOK_DELETE_001	Successful book deletion by Librarian	• Pre-condition: Create a book that has no associated borrowals. • Login as Librarian. • Navigate to 'Book Management'. • Select the book and click 'Delete'. • Confirm the deletion.	• System accepts the deletion. • A success message is displayed. • The book is removed from the book list.
Continued on next page			

Table 1.2 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_BOOK_DELETE_001	(Referential Integrity) Attempt to delete a book that is currently borrowed	• Pre-condition: A book exists and is part of an active borrowal record. • Login as Librarian. • Navigate to 'Book Management'. • Attempt to delete the borrowed book.	• System prevents the deletion. • An appropriate error message is displayed (e.g., "Cannot delete a book with active borrowals"). • The book is not deleted.
Sub-Section: Author Management (SRS 3.4)			
TC_AUTHOR_CREATE_001	Successful new author creation by Librarian	• Login as Librarian. • Navigate to 'Author Management'. • Click 'New Author'. • Fill the required 'Name' field with valid data. • Submit the form.	• A success message is displayed. • The new author appears in the authors list.
TC_AUTHOR_DELETE_001	(Referential Integrity) Attempt to delete an author linked to a book	• Pre-condition: An author is assigned to at least one book in the system. • Login as Librarian. • Navigate to 'Author Management'. • Attempt to delete the author.	• System prevents the deletion. • An error message is displayed (e.g., "Cannot delete author assigned to existing books"). • The author is not deleted.
Sub-Section: Genre Management (SRS 3.5)			
TC_GENRE_CREATE_001	Successful new genre creation by Librarian	• Login as Librarian. • Navigate to 'Genre Management'. • Click 'New Genre'. • Fill the required 'Name' and 'Description' fields with valid data. • Submit the form.	• A success message is displayed. • The new genre appears in the genres list.
Continued on next page			

Table 1.2 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_GENRE_DELETE_001	(Referential Integrity) Attempt to delete a genre linked to a book	- Pre-condition: A genre is assigned to at least one book in the system. - Login as Librarian. - Navigate to 'Genre Management'. - Attempt to delete the genre.	- System prevents the deletion. - An error message is displayed (e.g., "Cannot delete genre assigned to existing books"). - The genre is not deleted.
Sub-Section: User Management (SRS 3.7)			
TC_USER_CREATE_001	Successful new user (Member) creation by Librarian	- Login as Librarian. - Navigate to 'User Management'. - Click 'New User'. - Fill required fields (Name, Email, Password) with valid, unique data. Set 'isAdmin' flag to false. - Submit the form.	- A success message is displayed. - The new user appears in the user list.
TC_USER_CREATE_002	Attempt to create a user with an existing email	- Login as Librarian. - Navigate to 'User Management'. - Attempt to create a new user with an email address that is already in use.	- System rejects the creation. - An error message like "User already exists" is displayed.
TC_USER_READ_001	(RBAC) Verify Member cannot access User Management	Login as a Member.	'User Management' link/feature is not visible or is disabled. - Direct navigation to the user management URL is denied.
TC_USER_DELETE_001	(Referential Integrity) Attempt to delete a member with active borrowals	- Pre-condition: A member has one or more active (not returned) borrowal records. - Login as Librarian. - Navigate to 'User Management'. - Attempt to delete the member.	- System prevents the deletion. - An error message is displayed (e.g., "Cannot delete member with active borrowals"). - The user is not deleted.
Continued on next page			

Table 1.2 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
Sub-Section: Borrowal Management (SRS 3.6)			
TC_BORROW_CREATE1	Successful borrowal creation by Member	• Pre-condition: A book is available (isAvailable=true). • Login as Member. • Navigate to 'Borrowal Management'. • Click 'New Borrowal' and select an available book. • Submit the form.	• A new borrowal record is created with 'Borrowed' status. • A success message is displayed. • The book's availability status is updated to 'false'.
TC_BORROW_CREATE2	Attempt to borrow an unavailable book	• Pre-condition: A book is unavailable (isAvailable=false). • Login as Member. • Attempt to create a borrowal for the unavailable book.	• The system prevents the borrowal. • An appropriate error message is displayed (e.g., "Book is not available").
TC_BORROW_READ1	Verify Member can only see their own borrowal history	• Pre-condition: Multiple members have borrowal records. • Login as Member 1. • Navigate to 'Borrowal Management'.	• The list only displays borrowals where the memberId matches Member 1's ID. • Borrowals for other members are not visible.
TC_BORROW_UPDATE1	Successful borrowal update by Librarian (mark as returned)	• Pre-condition: An active borrowal exists. • Login as Librarian. • Navigate to 'Borrowal Management'. • Select the active borrowal and edit it. • Change the status to 'Returned'. • Save the changes.	• The borrowal record status is updated. • The associated book's 'isAvailable' status should ideally be updated to 'true'. • A success message is displayed.
Sub-Section: Review Management (SRS 3.8)			
Continued on next page			

Table 1.2 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_REVIEW_CREATE_SUCCESS	Successful review creation by Member	• Pre-condition: Based on assumed member capabilities if UI were present. • Login as Member. • Navigate to a specific book's detail page. • Submit a new review with a star rating and text.	• A success message is displayed. • The new review is visible on the book's page.
TC_REVIEW_CREATE_FAIL	Attempt to create a review with invalid data	• Login as Member. • Attempt to submit a review with a required field missing (e.g., star rating) or an invalid value (e.g., 6 stars).	• System rejects the submission. • An appropriate error message is displayed.
TC_REVIEW_DELETE_SUCCESS	Successful review deletion by Librarian	• Pre-condition: A review exists for a book. • Login as Librarian. • Navigate to the book or a review management section. • Select a review and delete it.	• The review is successfully deleted. • The review is no longer visible.

1.3 Specific Feature Testing

This section focuses on testing specific functionalities within the system that might not be fully covered by CRUD or Use Case tests, such as the Librarian Dashboard and detailed access controls for various entities. These tests ensure that individual features operate as designed and that role-based permissions are respected.

1.3.1 Dashboard (Librarian)

Verifies that only Librarians can access the dashboard and that it displays relevant overview statistics and charts.



Figure 1.1: Librarian Dashboard Overview

1.3.2 Book Management (Detailed)

Further tests on book CRUD operations, particularly focusing on UI element visibility and direct API access denial for members.

1.3.3 Borrowal Management (Detailed)

Additional tests to confirm successful borrowal requests, handling of unavailable books, and verification of role-based viewing of borrowal history.

1.3.4 User Management (by Librarian) (Detailed)

Detailed tests for Librarian's abilities to manage user accounts, including viewing all users, updating details, and successful deletion.

1.3.5 Review Management (Detailed)

Detailed tests on the new review feature, including addition and deletion by appropriate roles, and general visibility of reviews.

The detailed test cases for Specific Feature Testing are provided in:

Table 1.3: Specific Feature Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Sub-Section: Dashboard (Librarian)			
			Continued on next page

Table 1.3 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_DASH_001	Verify Librarian can access the Dashboard	• Login as Librarian. • Navigate to the Dashboard.	• User is successfully redirected to the Librarian dashboard at '/dashboard'. • Dashboard displays overview statistics and charts.
TC_DASH_002	Verify Member cannot access the Dashboard	• Login as a 'Member' user. • Attempt to navigate directly to the '/dashboard' URL.	• Access is denied. • User is redirected to the member landing page (e.g., '/books') or an error page.
Sub-Section: Book Management (CRUD)			
TC_BOOK_ADD_001	Successful new book creation by Librarian	• Login as Librarian. • Navigate to 'Book Management' (/books). • Click "New Book" button. • Fill in all required fields (Name, ISBN) with valid data. • Select existing Author and Genre. • Submit the form.	• A success message is displayed. • The new book record is created and appears in the book list.
TC_BOOK_ADD_002	Attempt to create a book with missing required fields	• Login as Librarian. • Navigate to 'Book Management'. • Open the "Add Book" form. • Leave a required field (e.g., Name) blank. • Submit the form.	• The system rejects the submission. • An error message indicating the required field is missing is displayed. • The book is not added to the database.
TC_BOOK_VIEW_001	Verify Member can view list of books and book details	• Login as Member. • Navigate to 'Book Management' (/books). • Click on a specific book in the list.	• Member can successfully view the list of all books. • Member can successfully view the details of a specific book.
Continued on next page			

Table 1.3 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_BOOK_UPD_001	Successful update of a book's details by Librarian	• Login as Librarian. • Navigate to 'Book Management'. • Select a book to edit. • Change a field (e.g., update the summary). • Save the changes.	• A success message is displayed. • The updated information is reflected correctly in the book's detail view.
TC_BOOK_DEL_001	Successful deletion of a book by Librarian	• Login as Librarian. • Navigate to 'Book Management'. • Select a book to delete. • Confirm the deletion in the dialog.	• A success message is displayed. • The book is removed from the book list.
TC_BOOK_ACCESS_001	Verify Member cannot access book CRUD operations	• Login as Member. • Navigate to 'Book Management'.	• "Add Book", "Edit Book", and "Delete Book" UI elements are not visible or are disabled. • Direct access to corresponding API endpoints is denied.
Sub-Section: Borrowal Management			
TC_BORW_ADD_001	Successful borrowal request by Member for an available book	• Login as Member. • Navigate to 'Borrowal Management' (/borrowals). • Click "New Borrowal". • Select a book that is marked as available. • Submit the form.	• A success message is displayed. • A new borrowal record is created with 'Borrowed' status and correct dates. • The new record appears in the member's borrowal history.
TC_BORW_ADD_002	Attempt to borrow a book that is not available	• Login as Member. • Navigate to 'Borrowal Management'. • Attempt to create a new borrowal for a book that is not available.	• The UI should clearly indicate the book is not available. • The system prevents the submission or displays a clear error message.
Continued on next page			

Table 1.3 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_BORW_VIEW_001	Verify Member can only view their own borrowal history	• Login as a Member who has borrowal records. • Navigate to 'Borrowal Management' (/borrowals).	• The list displays only the borrowals associated with the logged-in member's ID. • Records of other members are not visible.
TC_BORW_VIEW_002	Verify Librarian can view all borrowal records	• Login as Librarian. • Navigate to 'Borrowal Management'.	• A list of all borrowal records for all members is displayed.
TC_BORW_UPD_001	Successful update of borrowal status by Librarian	• Login as Librarian. • Navigate to 'Borrowal Management'. • Select an existing 'Borrowed' record. • Update its status to 'Returned'. • Save the change.	• A success message is displayed. • The record's status is correctly updated in the list.
Sub-Section: User Management (by Librarian)			
TC_USER_VIEW_001	Verify Librarian can view list of all users	• Login as Librarian. • Navigate to 'User Management' (/users).	• A list of all users (both Librarians and Members) is displayed correctly.
TC_USER_UPD_001	Successful update of a user's details by Librarian	• Login as Librarian. • Navigate to 'User Management'. • Select a user to edit. • Update a non-critical field (e.g., Phone number). • Save the changes.	• A success message is displayed. • The user's details are correctly updated.
TC_USER_DEL_001	Successful deletion of a user by Librarian	• Login as Librarian. • Navigate to 'User Management'. • Select a user account to delete. • Confirm the deletion.	• A success message is displayed. • The user is removed from the list of users. • The deleted user can no longer log in.
Sub-Section: Review Management (Assumed Admin UI)			
Continued on next page			

Table 1.3 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_REV_ADD_001	Successful addition of a review by Librarian (Admin action)	• Login as Librarian. • Navigate to a book's detail page. • Use an administrative function to add a review for the book. • Enter a star rating and review text. • Submit the review.	• A success message is displayed. • The new review appears in the list of reviews for that book.
TC_REV_VIEW_001	Verify any user can view reviews for a book	• As any logged-in user (or guest, if public). • Navigate to the detail page of a book that has reviews.	• The list of reviews for the book is displayed correctly.
TC_REV_DEL_001	Successful deletion of a review by Librarian	• Login as Librarian. • Navigate to a book with reviews. • Select a specific review to delete. • Confirm the deletion.	• A success message is displayed. • The review is removed from the book's list of reviews.

1.4 Use Case Testing

Use Case testing directly validates the main and alternative flows of key user interactions as defined in the SRS's Use Case section. This approach ensures that the system supports essential user tasks comprehensively.

1.4.1 UC-002: Add New Book (Librarian)

Verifies the complete process of a Librarian adding a new book, including successful scenarios and handling of missing required fields or backend validation failures.

1.4.2 UC-003: Borrow a Book (Member)

Ensures that members can successfully borrow available books, that the system prevents borrowing unavailable books, and that the borrowing form correctly auto-populates relevant data.

1.4.3 UC-005: View Borrowal History (Member)

Validates a member's ability to view their own borrowal history, including scenarios with empty history and ensuring proper role-based viewing (members see only theirs, librarians see all).

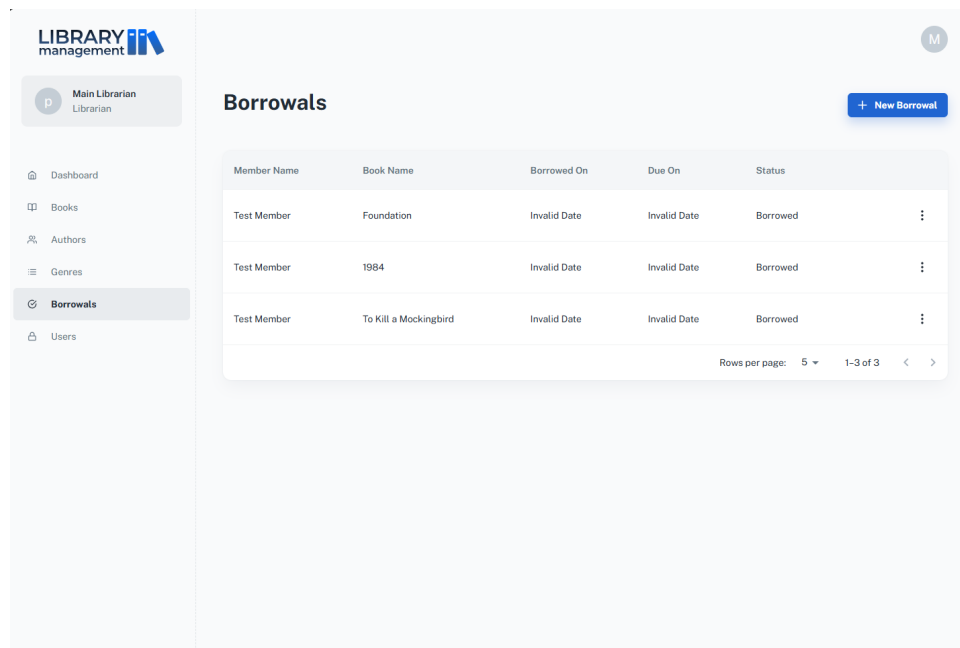


Figure 1.2: Member's Borrowal History View

The detailed test cases for Use Case Testing are provided in:

Table 1.4: Use Case Test Cases

Test Case ID	Test Title / Use Case	Test Steps	Expected Results
Sub-Section: UC-002: Add New Book (Librarian)			
TC_UC_BOOK_001	Successful addition of a new book (Main Flow)	- As a logged-in Librarian, navigate to the Book Management page ('/books'). - Click the "New Book" button. - Fill in all required fields (Name, ISBN) with valid data. - Select a valid Author and Genre from the provided lists. - Ensure "Is Available" is set to true. - Submit the form.	- The system creates a new book record via 'POST /api/book/add'. - A success message is displayed (e.g., "Book added"). - The "Add Book" form closes. - The new book appears in the list on the Book Management page.
Continued on next page			

Table 1.4 – continued from previous page

Test Case ID	Test Title / Use Case	Test Steps	Expected Results
TC_UC_BOOK_002	Attempt to add a book with missing required fields (Alternative Flow A1)	- As a logged-in Librarian, navigate to the "Add Book" form. - Fill in the form but leave a required field (e.g., Name) blank. - Submit the form.	- The system rejects the submission due to a validation error. - An error message is displayed, indicating the required field is missing. - The form remains open for correction. - The book is not added to the database.
TC_UC_BOOK_003	Attempt to add a book that fails backend validation (Alternative Flow A2)	- As a logged-in Librarian, navigate to the "Add Book" form. - Fill in all fields with data that is syntactically valid on the client-side but might trigger a server error (e.g., a duplicate ISBN if the server enforces uniqueness). - Submit the form.	- The system displays a generic error message from the API (e.g., "Something went wrong, please try again"). - The book is not added to the database.
Sub-Section: UC-003: Borrow a Book (Member)			
TC_UC_BORROW_001	Successful borrowing of an available book (Main Flow)	- As a logged-in Member, navigate to the Borrowal Management page ('/borrowals'). - Click the "New Borrowal" button. - Select a book that is marked as available from the list. - Verify that the Member field is correctly auto-populated. - Submit the form.	- A new borrowal record is created with "Borrowed" status via 'POST /api/borrowal/add'. - The availability of the book is updated to 'false'. - A success message is displayed. - The member's borrowal list is updated to show the new borrowal.
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Table 1.4 – continued from previous page

Test Case ID	Test Title / Use Case	Test Steps	Expected Results
TC_UC_BORROW_002	Attempt to borrow an unavailable book (Alternative Flow A1)	- As a logged-in Member, navigate to the "Add Borrowal" form. - Attempt to select a book that is not available (isAvailable=false).	- The UI should ideally prevent the selection of an unavailable book. - If selection is possible, the system should prevent submission or display an error message (e.g., "Book is not available") upon submission attempt. - No borrowal record is created.
TC_UC_BORROW_003	Verify correct data population in borrowal form	- As a logged-in Member, open the "Add Borrowal" form. - Observe the date fields.	- The Member field is correctly auto-populated with the current user's ID. - The Borrowed Date is auto-populated to the current date. - The Due Date is auto-populated to a future date (e.g., 14 days from current). - The initial status is correctly set (e.g., "Borrowed").
Sub-Section: UC-005: View Borrowal History (Member)			
Continued on next page			

Table 1.4 – continued from previous page

Test Case ID	Test Title / Use Case	Test Steps	Expected Results
TC_UC_HISTORY_001	View non-empty borrowal history (Main Flow)	• Log in as a Member who has previously borrowed one or more books. • Navigate to the Borrowal Management page ('/borrowals').	• The system fetches all borrowals and the client filters them. • A list/table is displayed showing *only* the records where the 'memberId' matches the logged-in Member's ID. • The displayed data (Book Name, Borrowed Date, Due Date, Status) is accurate for each record.
TC_UC_HISTORY_002	View empty borrowal history (Alternative Flow A1)	• Log in as a new Member with no borrowal records. • Navigate to the Borrowal Management page ('/borrowals').	• The system displays a message indicating there is no borrowal history (e.g., "No borrowals found").
TC_UC_HISTORY_003	Verify role-based view of borrowal history	• Prerequisite: There are borrowals from multiple members in the system. • Log in as a Librarian and navigate to the Borrowal Management page ('/borrowals'). • Log out and log in as a Member (who has at least one borrowal) and navigate to the same page.	• As the Librarian, all borrowal records for all members are visible. • As the Member, only their own borrowal history is visible.
Note: UC-001 (User Login) and UC-004 (Register New User) are detailed in the Authentication & Authorization test plan.			

1.5 State Transition Testing

State Transition testing focuses on how the system's state changes in response to specific events or actions, particularly for entities with defined lifecycle states such as Borrowal Records and Book Availability. This testing ensures that transitions between states occur correctly and that invalid transitions are prevented.

1.5.1 Borrowal Record Status

This section tests transitions for borrowal records, from 'None' to 'Borrowed', 'Borrowed' to 'Returned', 'Borrowed' to 'Overdue', and 'Overdue' to 'Returned'. It also includes tests for invalid transitions, such as reverting a 'Returned' status to 'Borrowed'.

1.5.2 Book Availability Status

These tests verify that a book's availability status ('Available' or 'Unavailable') changes correctly in response to borrowing and returning actions, and that borrowing is prevented for unavailable books.

1.5.3 User Session State

This sub-section examines the transitions of a user's session state, covering login, logout, accessing protected pages when logged out, and verifying session persistence after a page refresh.

The detailed test cases for State Transition Testing are provided in:

Table 1.5: State Transition Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Entity: Borrowal Record (Based on 'status' attribute)			
TC_STATE_BORROW_001	Valid Transition: (None) to 'Borrowed'	• Login as a Member. • Navigate to the Borrowal Management page. • Create a new borrowal record for an available book. • Submit the form.	• A new borrowal record is successfully created. • The status of the new record is set to "Borrowed". • A success message is displayed.
TC_STATE_BORROW_002	Valid Transition: 'Borrowed' to 'Returned'	• Precondition: A book is in 'Borrowed' status by a member. • Login as Librarian. • Navigate to Borrowal Management. • Locate the 'Borrowed' record. • Update the record's status to 'Returned' before its due date.	• The system accepts the update. • The borrowal record's status changes from "Borrowed" to "Returned". • A success message is displayed.

Continued on next page

Table 1.5 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_STATE_BORROW_001	Valid Transition: 'Borrowed' to 'Overdue'	• Precondition: A book is in 'Borrowed' status by a member. • Manually adjust system time or the record's due date to be in the past. • As a Librarian, view the borrowal record. (Or trigger a system check for overdue items).	• The borrowal record's status automatically (or upon view) changes from "Borrowed" to "Overdue". • The status is displayed correctly in the UI.
TC_STATE_BORROW_002	Valid Transition: 'Overdue' to 'Returned'	• Precondition: A book is in 'Overdue' status. • Login as Librarian. • Locate the 'Overdue' borrowal record. • Update the record's status to 'Returned'.	• The system accepts the update. • The borrowal record's status changes from "Overdue" to "Returned". • A success message is displayed.
TC_STATE_BORROW_003	Invalid Transition: 'Returned' to 'Borrowed'	• Precondition: A borrowal record has a 'Returned' status. • As a Librarian, attempt to edit the same record and change its status back to 'Borrowed'.	• The system should reject the change. • The UI should not provide an option for this transition, or it should result in an error if attempted via API. • An appropriate error message is displayed (e.g., "Cannot revert a returned item").
Entity: Book (Based on 'isAvailable' attribute)			
TC_STATE_BOOK_001	Valid Transition: 'Available' to 'Unavailable'	• Precondition: A book exists with 'isAvailable' status set to true. • As a Member, create a new borrowal for that specific book. • After the borrowal is confirmed, view the details of that book.	• The borrowal transaction is successful. • The 'isAvailable' status of the book is updated to false. • The book's listing correctly shows it as unavailable.
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Table 1.5 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_STATE_BOOK_002	Valid Transition: 'Unavailable' to 'Available'	• Precondition: A book has 'isAvailable' status set to false due to being borrowed. • As a Librarian, locate the corresponding borrowal record. • Update the borrowal record status to 'Returned'. • View the details of the book again.	• The 'isAvailable' status of the book is updated to true. • The book's listing correctly shows it as available for borrowing.
TC_STATE_BOOK_003	Invalid Transition: Attempt to borrow an 'Unavailable' book	• Precondition: A book has 'isAvailable' status set to false. • As a Member, attempt to create a new borrowal for that specific book.	• The system prevents the borrowal request from being submitted. • An appropriate error message is displayed (e.g., "Book is currently unavailable"). • No new borrowal record is created.
Entity: User Session (Based on login/logout actions)			
TC_STATE_SESSION_001	Valid Transition: 'Logged-Out' to 'Logged-In'	• Start from a logged-out state (e.g., in an incognito window). • Navigate to the login page. • Enter valid credentials for a 'Member'. • Click 'Login'.	• User session is established. • User is redirected to the member landing page (e.g., /books). • User can access member-specific pages.
TC_STATE_SESSION_002	Valid Transition: 'Logged-In' to 'Logged-Out'	• Precondition: A user is logged in. • Click the 'Logout' button/link.	• The user's session is terminated. • User is redirected to the login page.
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Table 1.5 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_STATE_SESSION05	Invalid Transition: Accessing protected page when 'Logged-Out'	• Start from a logged-out state. • Attempt to directly navigate to a protected URL (e.g., /borrowals, /users).	• Access to the protected page is denied. • User is redirected to the login page.
TC_STATE_SESSION06	State Persistence: Verify session remains 'Logged-In' after page refresh	• Login as any user. • Navigate to any page other than the login page. • Refresh the web browser page.	• The user remains logged in. • The user stays on the same page. • No redirection to the login page occurs.